

## EXERCISE SHEET 10

### GROUP AND RING THEORY

#### MAIN QUESTIONS - TO BE DONE BEFORE THE SEMINAR

**10.1.** Let  $K$  be a field. Use Hilbert's basis theorem to show that the subring  $K[x^2, x^3]$  of  $K[x]$  is Noetherian.

**10.2.** Show that if  $A$  and  $B$  are nilpotent right (or left) ideals, their sum  $A + B$  is also nilpotent.

**10.3.** Let  $R = \mathbb{Z}[x, y]/I$ , where  $I = (x^2 - 1, xy - y, y^2 - 2y)$ ; recall also that the variables  $x, y$  commute, so  $xy = yx$ . Show that

- (a)  $R$  has no non-zero nilpotent elements;
- (b)  $R$  is not the direct sum of minimal ideals.

#### EXTRA QUESTIONS

**10.4.** Consider

$$R = \begin{pmatrix} \mathbb{Q} & 0 \\ \mathbb{R} & \mathbb{R} \end{pmatrix} = \left\{ \begin{pmatrix} a & 0 \\ c & d \end{pmatrix} \mid a \in \mathbb{Q}, c, d \in \mathbb{R} \right\}.$$

Prove that  $R$  is left Noetherian and left Artinian, but neither right Noetherian nor right Artinian.

**10.5.** The game of Chomp starts with a half-infinite chessboard with squares labelled  $(n, m)$  for integers  $n, m \geq 0$  with a cookie on each square. Two players take turns choosing a cookie (i.e., not an empty square) and eating all the cookies on that square and the ones that are not to the left or below it; that is, if a player chooses the cookie on  $(n, m)$ , then they eat all the cookies on  $(n', m')$  for  $n' \geq n$  and  $m' \geq m$ . The cookie on  $(0, 0)$  is poisoned, and the player who eats it immediately loses. Prove that any game of Chomp ends after finitely many moves.

**10.6.** Recall that an element  $u$  of a ring  $R$  is called a *unit* if there exists a  $v \in R$  such that  $uv = vu = 1$ . Kaplansky's unit conjecture (dating back to 1940) says that if  $F$  is a field and  $G$  is a group without elements of finite order, then the only units of the group algebra  $F[G]$  are the trivial units, that is, the non-zero scalar multiples of group elements. A counterexample to this conjecture was discovered in 2021 by Giles Gardam for fields of characteristic 2, then by Alan Murray later in 2021 for other fields of positive characteristic, and very recently the characteristic zero case was disproved by Gardam in 2024.

Let  $G = D_3 = \langle x, y \mid x^3 = y^2 = 1, yxy = x^{-1} \rangle$  be the dihedral group of order 6 and consider the group algebra

$$\mathbb{C}[G] = \left\{ \sum c_0 + c_1x + c_2x^2 + c_3y + c_4xy + c_5x^2y \mid c_0, c_1, c_2, c_3, c_4, c_5 \in \mathbb{C} \right\}.$$

Show that  $z := (y - 1)x(1 + y)$  squares to zero, and hence find a non-trivial unit in  $\mathbb{C}[G]$ .